Tabish Izhar

Software Engineer focused on AI & Machine Learning Systems

Pestalozzistraße 9a, 96052 Bamberg, Germany | +49 1522 9106841 | tabishizhar9@gmail.com
github.com/tabishizhar9 | linkedin.com/in/tabishizhar9 | tabishizhar9.github.io

Profile

Software engineer with about three years of industry experience across game development, backend development and IT support, now pursuing an M.Sc. in International Software Systems Science at the University of Bamberg. Moving into AI/ML engineering, with coursework in machine learning, deep learning for NLP, data streams and cloud-native systems (Docker, Kubernetes, microservices). Experienced in Scrum teams and in taking features from idea to release.

Education

M.Sc. International Software Systems Science (ISoSySc) Apr 2024 – Present
Otto-Friedrich-Universität Bamberg *Bamberg, Germany*

- **Service-Oriented Architecture:** Built a Java web service locally and exposed it through REST APIs using Jersey; studied microservice architecture, Docker and Kubernetes.
- **Data Analytics & Machine Learning (ADAML):** Linear regression, logistic regression and k-nearest neighbours (KNN).
- **Deep Learning for NLP:** Tokenization, text representation and neural approaches to natural language processing.
- **Project – Dialogue Systems:** Built the Natural Language Understanding (NLU) component of a chatbot (see Projects).
- **Data Stream Processing:** Concepts and techniques for processing continuous data streams.
- **Evidence-Based Software Engineering:** Research paper on security vulnerabilities in AI-generated code (see Papers).
- **C++ Programming (DT-CPP-M):** Solved a large set of C++ assignments; every solution passed all automated test cases.
- **Databases (DBCPU-M):** SQL query practice; PostgreSQL work on the Enron dataset.

B.Sc. Computer Science 2017 – 2021
National College of Business Administration & Economics (NCBAE) *Pakistan*

- Teaching Assistant for Programming Fundamentals, Digital Logic Design and Discrete Structures.

Professional Experience

Software Engineer Oct 2022 – Apr 2024
Gaminations *Lahore, Pakistan*

- Worked on full-cycle development using C++, Python, JavaScript, PHP and MySQL for high-performance, feature-rich game applications.
- Developed automation scripts in Shell and Python to streamline Git operations and validate JSON-based game levels against acceptance criteria.
- Contributed to gameplay design with FairyGUI, Cocos2d-x and Cocos Creator, implementing features such as PowerUps, PopUps and tutorials to increase user engagement.
- Maintained and improved backend logic in PHP to meet changing product requirements.
- Managed Hotfix and Release branches for clean deployment workflows.
- Designed and implemented UI/UX elements using FGUI and JSON configurations for new features.
- Monitored app performance through Firebase Console and Google Play Console; tracked and reported bugs in Atlassian Jira.
- Worked in a Scrum team using Jira, Confluence and SourceTree for tasks, sprints and collaboration.

Dental CAD/CAM & IT Support Mar 2021 – Oct 2022
ClearPath Orthodontics *Lahore, Pakistan*

- Worked with 3Shape OrthoAnalyzer and Dent One for orthodontic case analysis and digital workflows.
- Provided IT support for hardware, software and network troubleshooting within the department.
- Managed the department's IT support for one month, handling desktop and network troubleshooting.
- Collaborated with clinical teams to optimise digital dental processes.

Projects

Linear Regression from Scratch | *Python, NumPy, FastAPI, Docker, Render*
github.com/tabishizhar9/linear-regression-from-scratch

- Implemented the cost function, gradients and gradient descent myself in NumPy; trained on the California Housing dataset using median income as the single feature.
- Cleaned the data by removing rows where the house price was capped at $500,000.
- Compared the result with scikit-learn's `LinearRegression`; the final cost was almost the same.
- Served the model through a FastAPI endpoint, containerised it with Docker and deployed it live on Render.

Rule-Based Chatbot – Dialogue Systems Project | *Python*
github.com/tabishizhar9/Rule-Based-Chatbot

- Master's project: a full dialogue system that helps users find a restaurant in Cambridge, UK.
- Built the Natural Language Understanding (NLU) component: regular-expression intent classification for 13 dialogue intents and slot filling for cuisine (about 90 types), price range and area, with several slots extracted from one query.
- NLU output feeds a modular pipeline: dialogue state update, database query on a Cambridge restaurant database, and template-based response generation.
- Validated with manually designed test cases covering all 13 intents, multi-slot queries, alternative requests and dialogue restarts.
- Currently exploring ML-based NLU, transformer models (BERT), LLMs and hybrid rule-based + ML approaches to improve accuracy and handle more varied queries.

Papers & Reports

The AI Security Showdown 2026
Evidence-Based Software Engineering, University of Bamberg – co-authored with Muhammad Usman

- Empirical study of the security of 883,140 AI-generated C programs from the FormAI-v2 dataset, produced by nine AI models and labelled with formal verification results (ESBMC model checker).
- Compared models by vulnerability density (bugs per 1,000 lines of code): GPT-4o-mini lowest (7.58), Gemma-7B highest (13.84).
- Found a weak but statistically significant negative correlation between cyclomatic complexity and vulnerability presence ($r = -0.0437$, $p < 0.05$).
- Analysis done in Python on Google Colab, with correlation tests and visualisations.

A Rule-Based Natural Language Understanding Module for a Domain-Constrained Restaurant Dialogue System Feb 2026
Project report, Master Project Dialogue Systems, University of Bamberg (supervisor: Dr. Stefan Ultes)

- Describes the design, testing and limitations of the chatbot's rule-based NLU module, and future directions such as statistical or neural intent classification, synonym handling and multi-intent detection.

Certifications & Ongoing Learning

- Machine Learning Specialization – DeepLearning.AI / Stanford (Coursera), in progress
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Technical Skills

Languages: Python, C++, JavaScript, PHP, Java, SQL, Shell
ML / Data: Linear & logistic regression, KNN, gradient descent, NumPy, scikit-learn, NLP basics (tokenization), regular expressions, correlation analysis, Jupyter, Google Colab
Backend / Web: FastAPI, REST APIs, Jersey, MySQL, PostgreSQL, MongoDB (coursework)
DevOps / Cloud: Docker, Kubernetes (coursework), microservices, Git, branching & release management, Render
Game Dev: Cocos2d-x, Cocos Creator, FairyGUI (FGUI)
Monitoring: Firebase Console, Google Play Console
Tools: Jira, Confluence, SourceTree
Domain tools: 3Shape OrthoAnalyzer, Dent One

Languages

English – Fluent | Urdu – Native | German – A2